

Lab Work: File Handling in Python

General Instructions for Students

Constraints

- Use **functions** to modularize the program.
 - Use **file handling** (open(), read(), write(), append()).
 - Use **lists/dictionaries** wherever appropriate.
 - Do **not use databases** or external libraries.
 - Implement **menu-driven execution using a while loop**.
 - Ensure that updates are reflected in the original file.
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1. Employee Payroll Management System

Problem Statement

A company stores employee details in a text file named employees.txt.

File Format

```
EMP101,Anuj,45000
EMP102,Rahul,52000
EMP103,Priya,38000
EMP104,Neha,61000
EMP105,Amit,29000
EMP106,Sneha,55000
EMP107,Karan,47000
EMP108,Pooja,72000
EMP109,Rohit,33000
EMP110,Anjali,68000
```

Requirements

Create a menu-driven program to:

1. Display all employee records.
2. Search employee details using Employee ID.
3. Calculate the average salary.
4. Find the highest-paid and lowest-paid employee.
5. Display employees earning above ₹50,000.
6. Add a new employee record to the file.
7. Generate salary categories:
 - High (₹60,000 and above)
 - Medium (₹40,000–₹59,999)
 - Low (Below ₹40,000)

2. Library Book Issue System

Problem Statement

A library stores book information in books.txt.

File Format

B101,Python Basics,5
B102,Java Programming,2
B103,Data Science,0
B104,DBMS,3
B105,Machine Learning,1
B106,Operating Systems,4
B107,Networking,2
B108,Cyber Security,6
B109,Cloud Computing,0
B110,Web Development,3

Requirements

Develop a program to:

1. Display all books.
2. Search a book using Book ID.
3. Issue a book (decrease quantity by 1).
4. Return a book (increase quantity by 1).
5. Display unavailable books.
6. Display books requiring restocking (copies < 2).
7. Update the file after every issue/return operation.

3. Student Result Processing System

Problem Statement

Student marks are stored in results.txt.

File Format

S101,Anuj,85
S102,Rahul,72
S103,Priya,96
S104,Neha,68
S105,Amit,39
S106,Sneha,54
S107,Karan,91
S108,Pooja,78
S109,Rohit,47
S110,Anjali,88

Requirements

Write a program to:

1. Display all student records.
 2. Search a student using Student ID.
 3. Find topper and lowest scorer.
 4. Calculate class average.
 5. Count pass and fail students.
 6. Generate grades:
 - A (90+)
 - B (75–89)
 - C (40–74)
 - F (<40)
 7. Write grade reports into a new file named grades.txt.
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4. Mobile Contact Directory System

Problem Statement

Contacts are stored in contacts.txt.

File Format

Anuj,9876543210
Rahul,9876543211
Priya,9876543212
Neha,9876543213
Amit,9876543214
Sneha,9876543215
Karan,9876543216
Pooja,9876543217
Rohit,9876543218
Anjali,9876543219

Requirements

Create a menu-driven application to:

1. Display all contacts.
 2. Search a contact by name.
 3. Add a new contact.
 4. Update an existing contact number.
 5. Delete a contact.
 6. Display contacts whose names start with a vowel.
 7. Save all modifications back to the file.
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5. Daily Expense Tracker and Report Generator

Problem Statement

Daily expenses are recorded in expenses.txt.

File Format

Food,450
Travel,300
Shopping,1200
Electricity,850
Internet,700
Entertainment,600
Medicine,400
Education,1500
Fuel,900
Miscellaneous,250

Requirements

Develop a program to:

1. Display all expenses.
 2. Calculate total expenditure.
 3. Find the category with highest and lowest spending.
 4. Display expenses greater than ₹800.
 5. Add a new expense category.
 6. Update an existing expense amount.
 7. Generate a summary report in report.txt containing:
 - Total Expenses
 - Highest Expense Category
 - Lowest Expense Category
 - Categories spending more than ₹800
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